



Mohammad Usaid
Civil Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 18/07/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	Core: 176 Total: 212

Completed Minor in Artificial Intelligence and Data Science at C-MInDS, IIT Bombay

PROFESSIONAL EXPERIENCE

Planning Engineer | Godrej Properties [May'25-Jul'25]

Transformed manual project management into data centric systems, enabling proactive risk mitigation & higher efficiency

- Automated trackers of 8 buildings, cutting reporting time by 40% & boosting data accuracy for better resource allocation
- Built an integrated & dynamic dashboard from trackers, boosting budget forecast accuracy by 20% with real time insights
- Developed a Python script to auto-generate vendor workflow diagrams, cutting coordination errors for 15+ contractors

AI/ML Research Intern | Research School, ICASSD [Jun'25-Present]

Guide: Prof. Apurva Narayan — Developing a ML model for integrated crop and energy yield prediction in Agro-PV systems

- Building a ConvLSTM model, leveraging STD-MAE framework to improve accuracy by 15% over baseline models
- Optimizing model training time by an anticipated 20% & enhancing performance by implementing STD-MAE framework

Data Analyst | NoQs Digital [Dec'23-Jan'24]

Received LOR & Star Intern Award for immaculate problem-solving abilities & displaying strong work ethics

- Extracted insights by analyzing datasets of 1000+ tasks, 50+ employee performance, and task sprint planning
- Automated dynamic dashboards in Excel for data-driven workload analysis, cutting man-hours by 2 hours/day

KEY PROJECTS

Hydro-Climatological System Dynamics | Research Project [May'25-Present]

Guide: Prof. Subimal Ghosh — Received Letter of Recommendation for exemplary work

- Applied causal inference & MI to a 6 variable system to map the complex soil-water dynamics for the Indian region
- Synthesizing findings to deliver a comprehensive diagnosis of drought propagation & soil moisture retention dynamics

Traffic Signal Optimization | Research Project [Aug'25-Present]

Guide: Prof. Archak Mittal — An AI-powered system for urban traffic optimization

- Architecting a 3-stage traffic optimization pipeline designed to reduce simulated intersection delays by 15–25%
- Developing a ResNeXt based vehicle density estimator, targeting to achieve MAE <1.5 per frame even under occlusion
- Integrating a pre-trained STD-MAE forecaster with a RL controller to reduce network wide traffic delays by over 20%

RAG-based Chatbot | OpenAI Open Model Hackathon [Aug'25-Sep'25]

- Built a RAG-based chatbot for conversational QA and quick information retrieval on PDFs and YouTube videos
- Enabled memory system to remember previous messages as context for new queries, making responses history-aware
- Integrated an agentic architecture to understand query context to route queries between RAG and web search

Defect Detection in Industrial Equipment | SOC — WnCC, IIT Bombay [May'24-Jul'24]

- Built a custom CNN leveraging a ResNet architecture to classify industrial equipment as defective vs. non-defective
- Employed data augmentation techniques and optimization to harden generalization, pushing test accuracy to 97.7%
- Deployed a pipeline with webcam I/O & on-device inference, enabling instant defect flags & production ready latency

POSITION OF RESPONSIBILITY

Department Placement Coordinator | IIT Bombay Placement Cell [Jul'25–Present]

Assisting in Placement of 2100+ students by working as a liaison between the placement office, IIT Bombay

- Elected to represent 151 students in Placement Team, contributing to its success by planning 15 upskilling activities
- Streamlined the company pitching with personalized preferences for 151 students, maximizing placement opportunities

ACHIEVEMENT & EXTRA-CURRICULARS

Mentorship	Mentored 5 students on GenAI in SOS'24, providing guidance & support to overcome challenges
Scholastic	- Secured 97.89%tile in JEE Advanced & 99.56%tile in JEE Mains — No of Candidates: 1.1M+ - Obtained 9.32 SPI in 3rd semester Awarded with an AA grade (perfect 10) in 5 civil courses
Competition	- Represented IIT Bombay at SDC'25 held in UC Berkeley, USA, securing 15th place overall - Among top 10 teams in the Pan IIT Solve The Case competition of SARCATHON'23 - Secured spot in Top 5 teams of Fetch.AI Hackathon, AI-based competition, in Techfest 2023-24
Technical	- Undertaken the WnCC AlgoNinjas Bootcamp'24, mastering Data Structures and Algorithms - Completed courses: Python for Data Science, Big Data Handling in Excel & SQL (Learners' Space'23)
Other	- Skills: Python, C++, SQL, Machine Learning Frameworks, Macro, Excel, CSS, JavaScript - Secured 1st position in cricket at the Inter-Hostel General Championship, representing Hostel 2 - Represented H2 in Institute Crossy GC, covering a distance of 4.5 km and secured 1st position - Organized a 15 km Cyclothon with 1,700+ participants, raising awareness on healthy living - Completed year-long professional training in YOGA, National Sports Organization at IITB